

Hippocampal LFP Decoding of Odor Sequence Memory

Predicting InSeq/OutSeq and Odor Identity from Rat Brain Waves Using
Generalized Linear Models

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About the Experiment

Behavioral Trial Workflow

- 1 Poke-In**
Rat enters port (anchors trial onset)
- 2 Odor Delivery**
1 of 5 odors presented (A, B, C, D, E)
- 3 Decision & Poke-Out**
Evaluates sequence; exits (~1.3s hold duration)

Sequence Protocol & Probes

IN-SEQUENCE TRIALS (InSeq ~ 87%)



Standard repeating sequence structure

OUT-OF-SEQUENCE PROBES (OutSeq ~13%)



One of the odors is out of sequence

RECORDING TARGET & DATASET

Hippocampal ensemble LFPs decoding nonspatial memory (5 rats)

Source Dataset: Hippocampal ensembles represent sequential relationships among an extended sequence of nonspatial events

About the Experiment

IN-SEQUENCE (EXPECTED)

ABCDE

HOLD POKE

≥ 1.0 seconds

Rat stays in port to signal correct sequence order.

OUT-OF-SEQUENCE (VIOLATION)

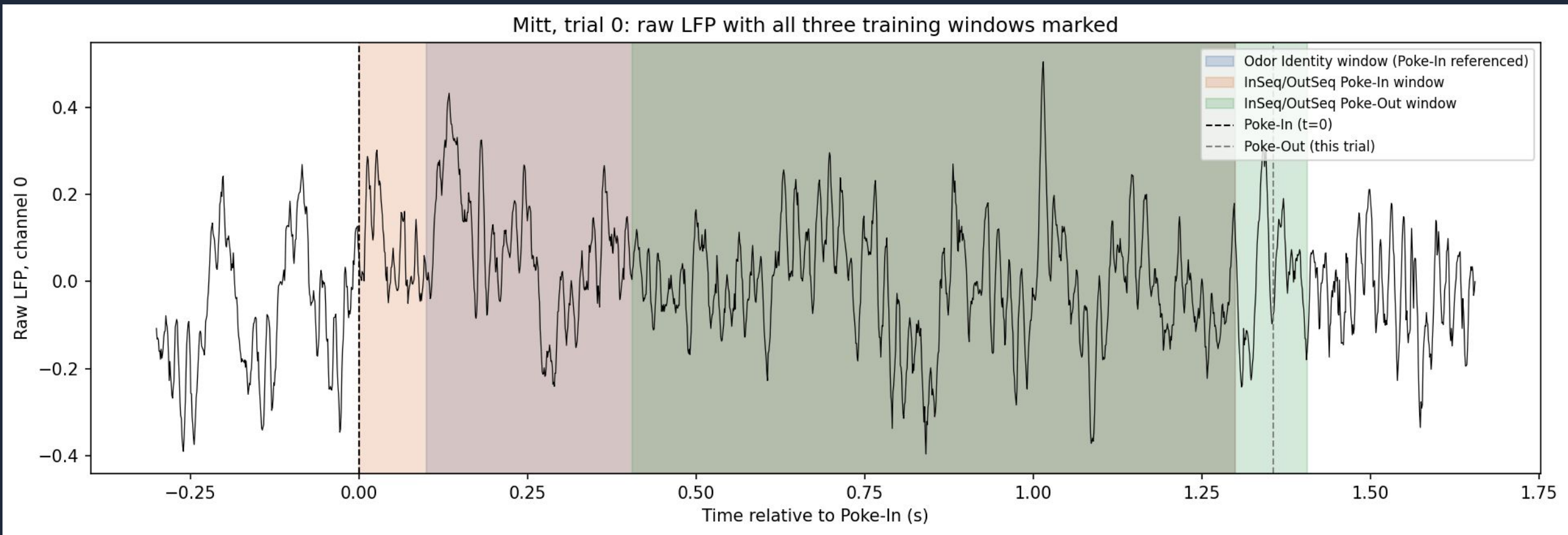
ABD (Wrong Odor!)

EXIT EARLY

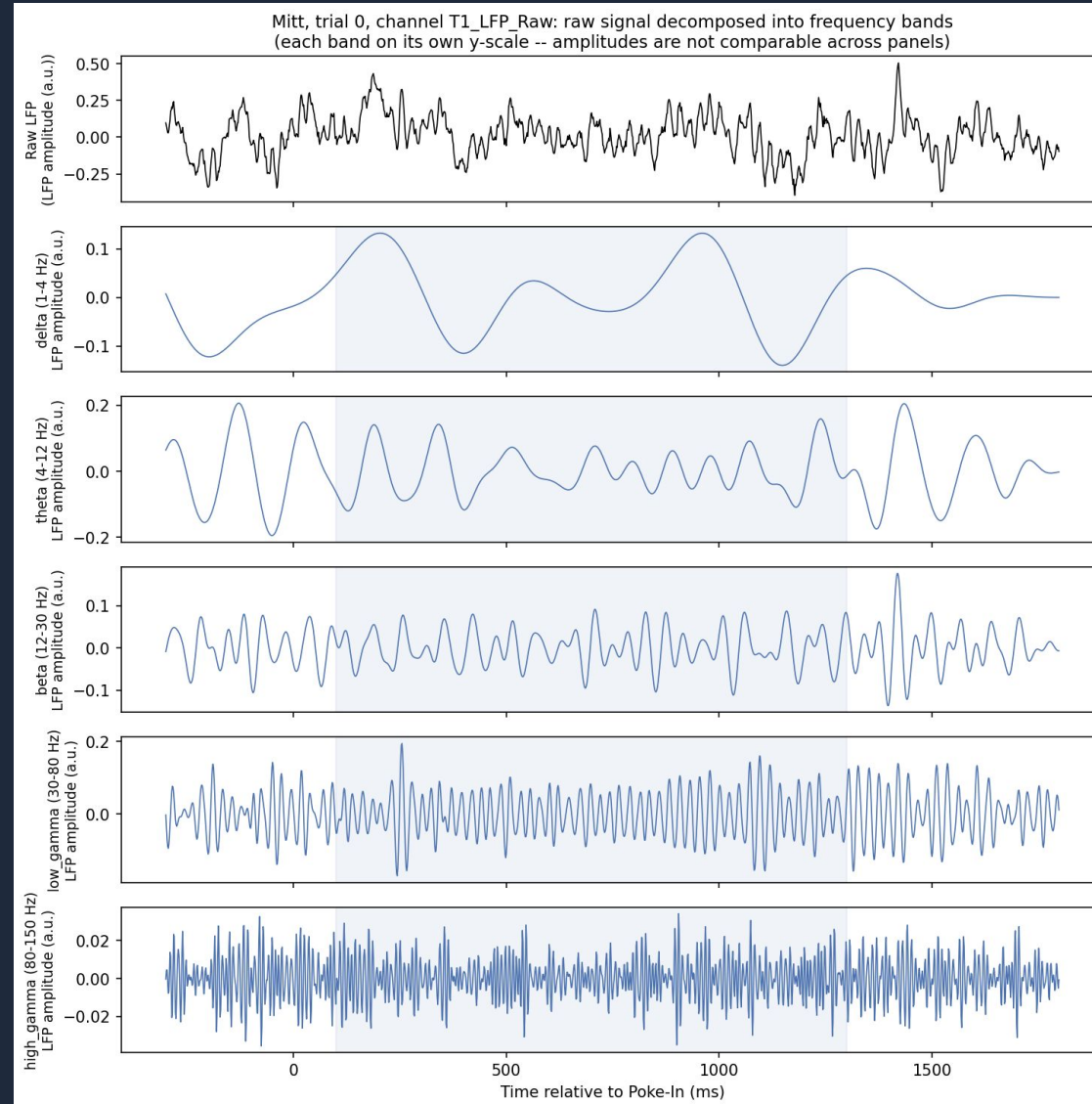
< 1.0 seconds

Rat withdraws quickly to signal incorrect sequence order..

What is Local Field Potential (LFP)



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Model Architecture & Decoding Pipeline

01. MODEL & INPUT FEATURES

GLM Architecture & Features

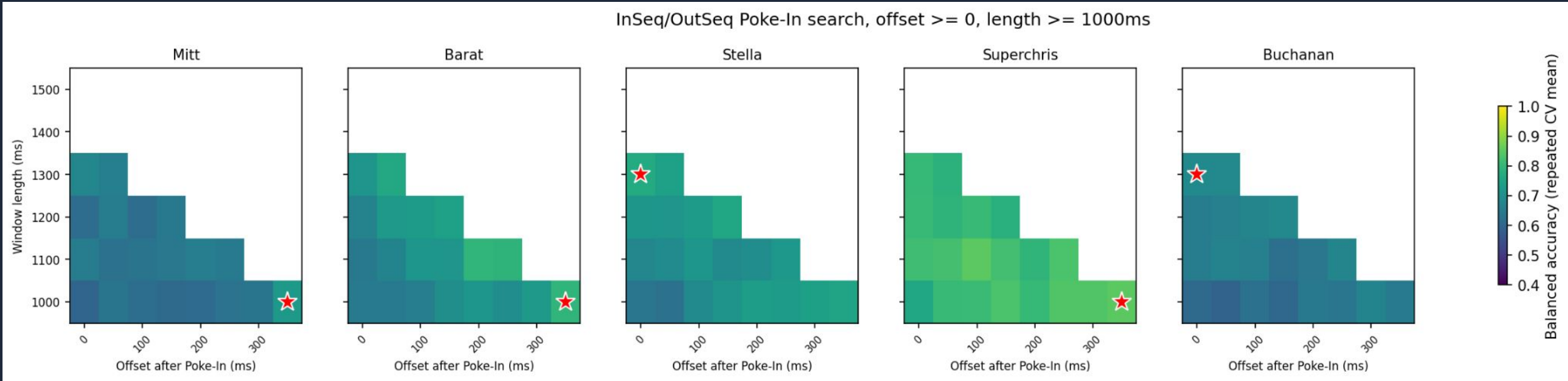
- **Ridge GLM:** Logistic regression with L2 regularization
- **20 Dimensions:** 5 frequency bands × 4 channels
- **Individual Training:** Models trained per rat on their own LFP data
- **Shared Window:** Same sliding window across all subjects (detailed later)

02. TARGETS & EVALUATION

Decoding & Mechanism Analysis

- **Targets:** Sequence memory (InSeq vs. OutSeq) & Odor identity
- **Validation:** Stratified CV evaluated via Accuracy & ROC-AUC
- **Interpretation:** Beta coefficients (β) reveal key brainwaves

Sliding Window



InSeq / OutSeq Decoding Performance – Poke-In

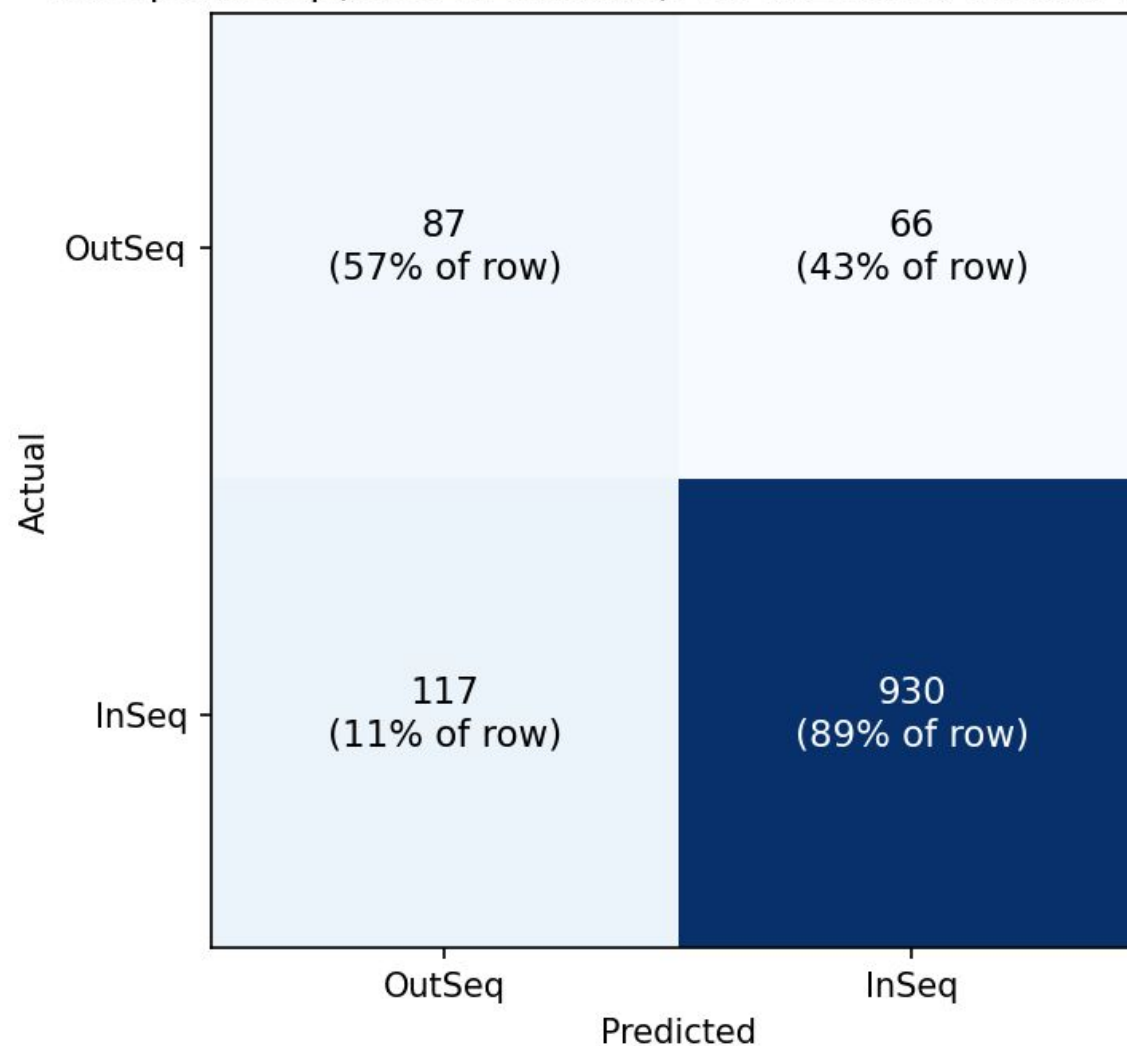
01. WINNING WINDOW & METRICS

InSeq/OutSeq Poke-In Analysis (0 to +1300 ms)

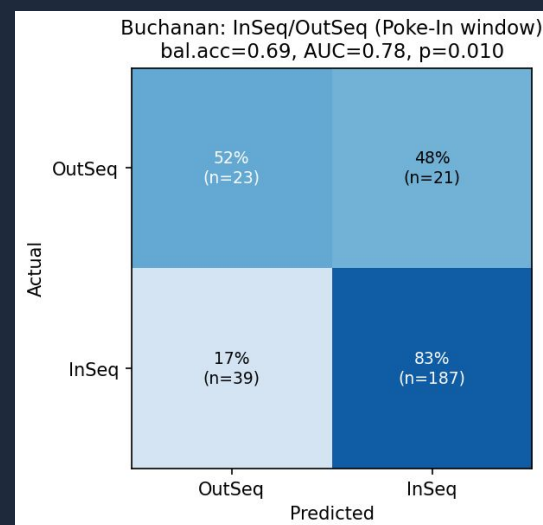
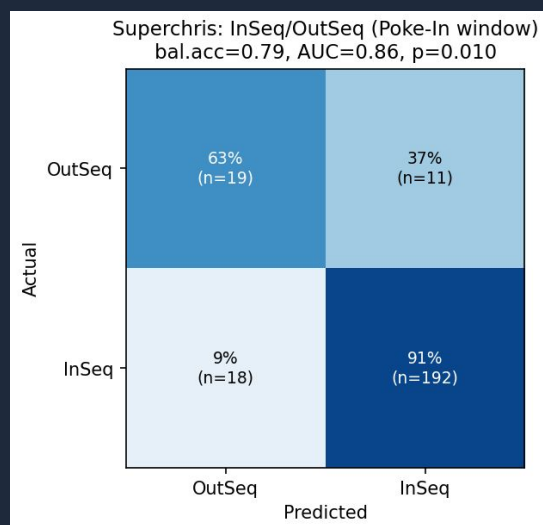
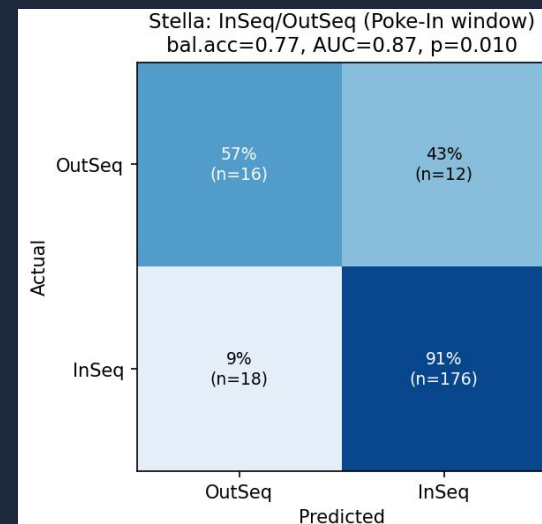
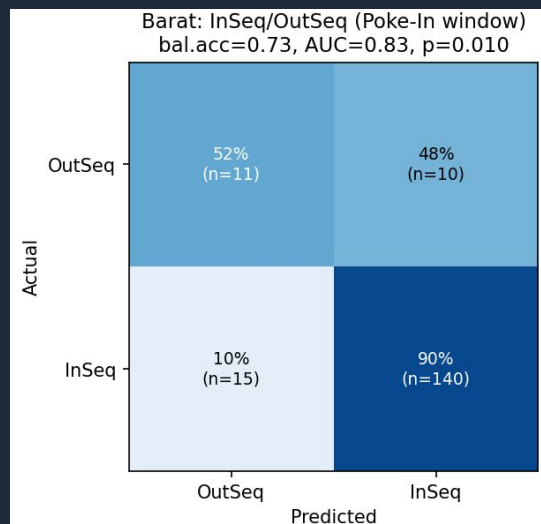
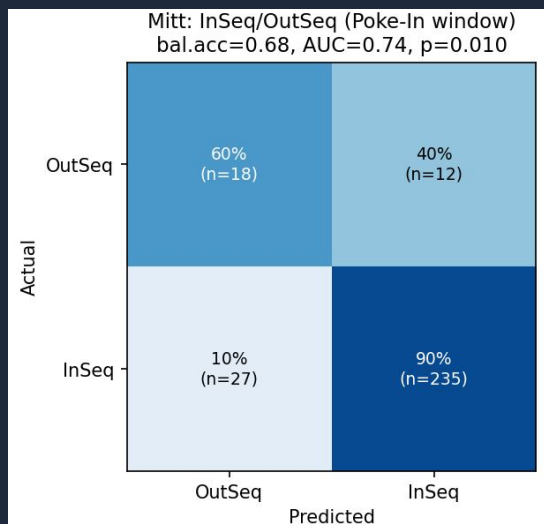
- 0 ms is the Poke-In event (odor onset / trial start)

AUC & Decoding Performance: **0.808**

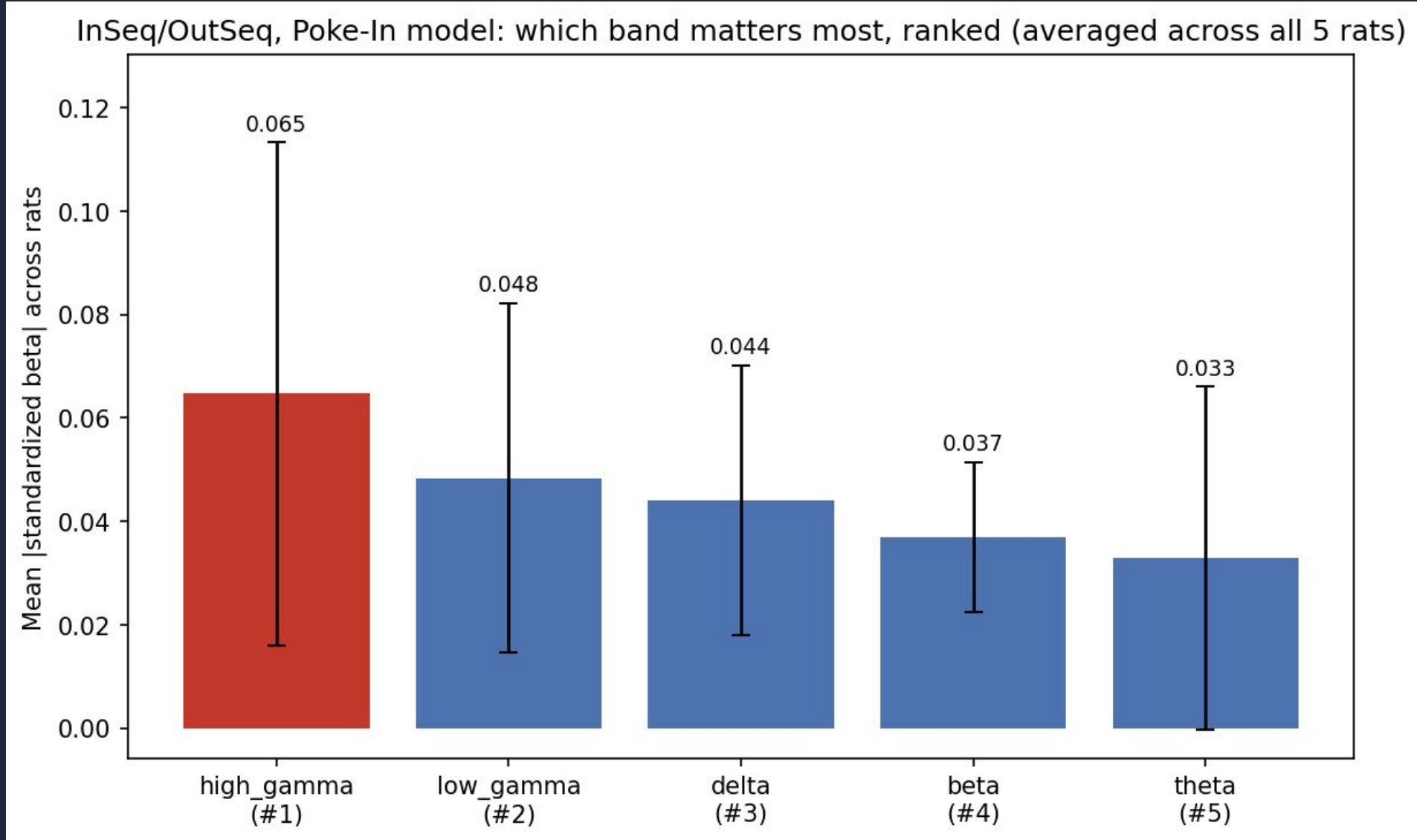
InSeq/OutSeq (Poke-In window): TOTAL trials, all rats combined



InSeq / OutSeq Decoding Performance – Poke-In



InSeq/OutSeq Poke-In Frequency Band



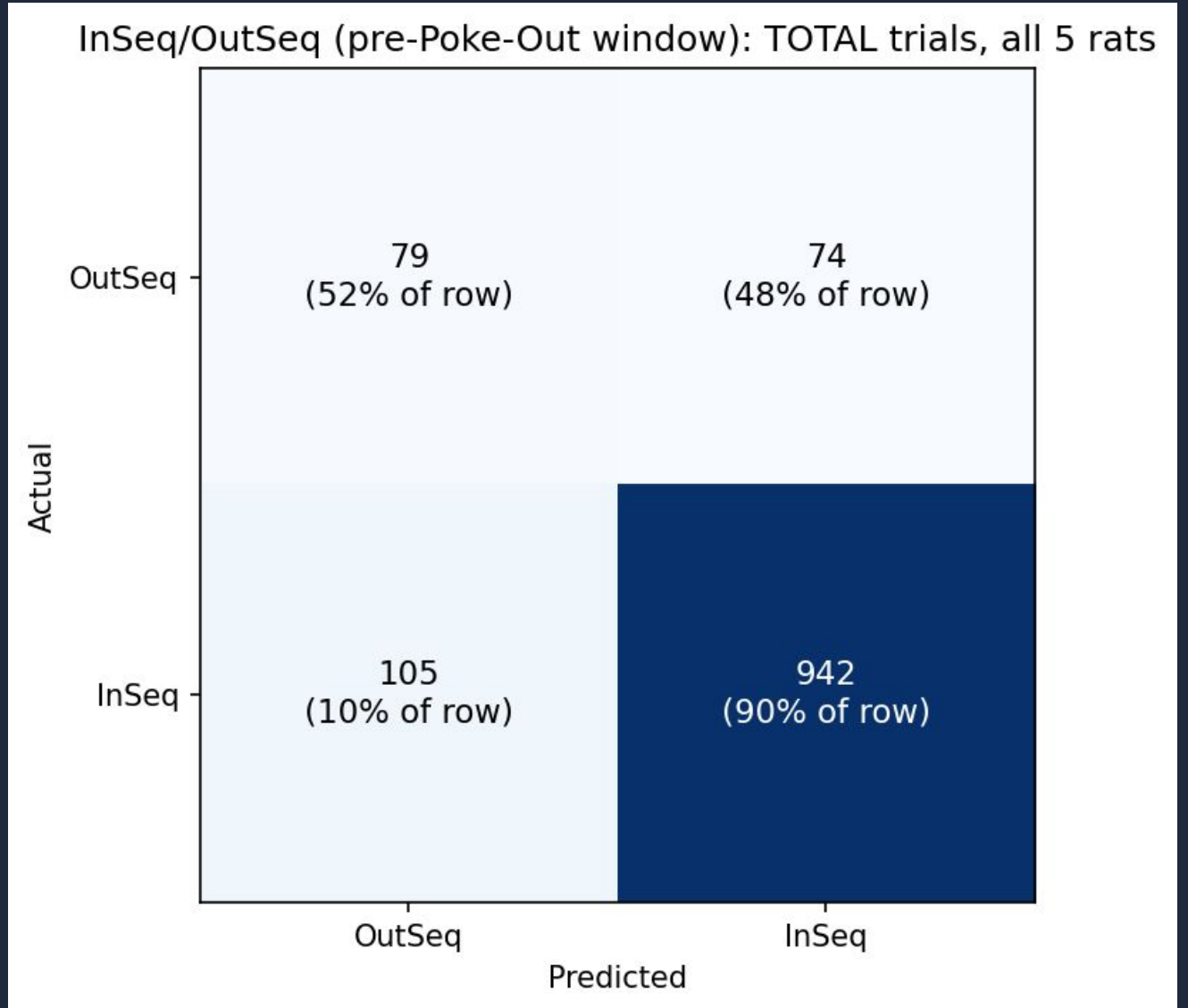
InSeq / OutSeq Decoding Performance – Poke-Out

01. WINNING WINDOW & METRICS

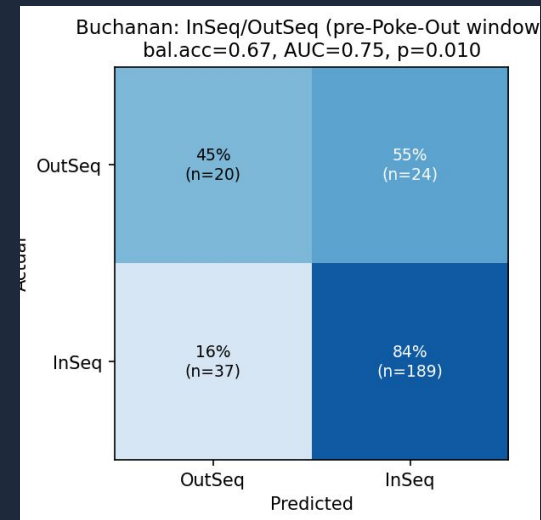
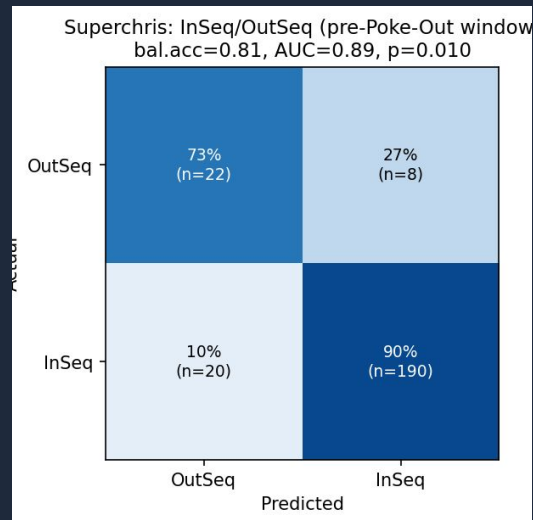
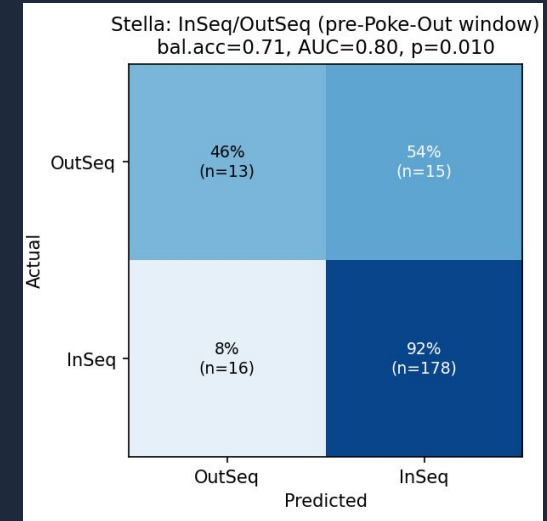
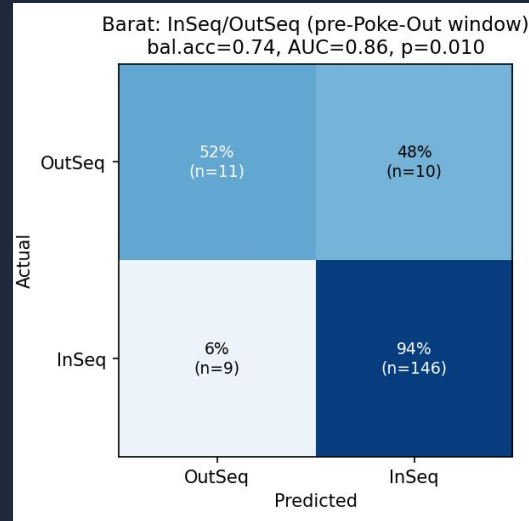
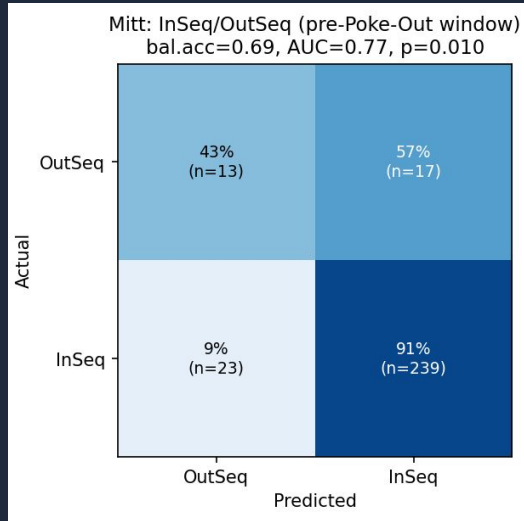
InSeq/OutSeq Poke-Out Analysis (-950 to +50ms)

- 0 ms is the Poke-Out event (rat decision)

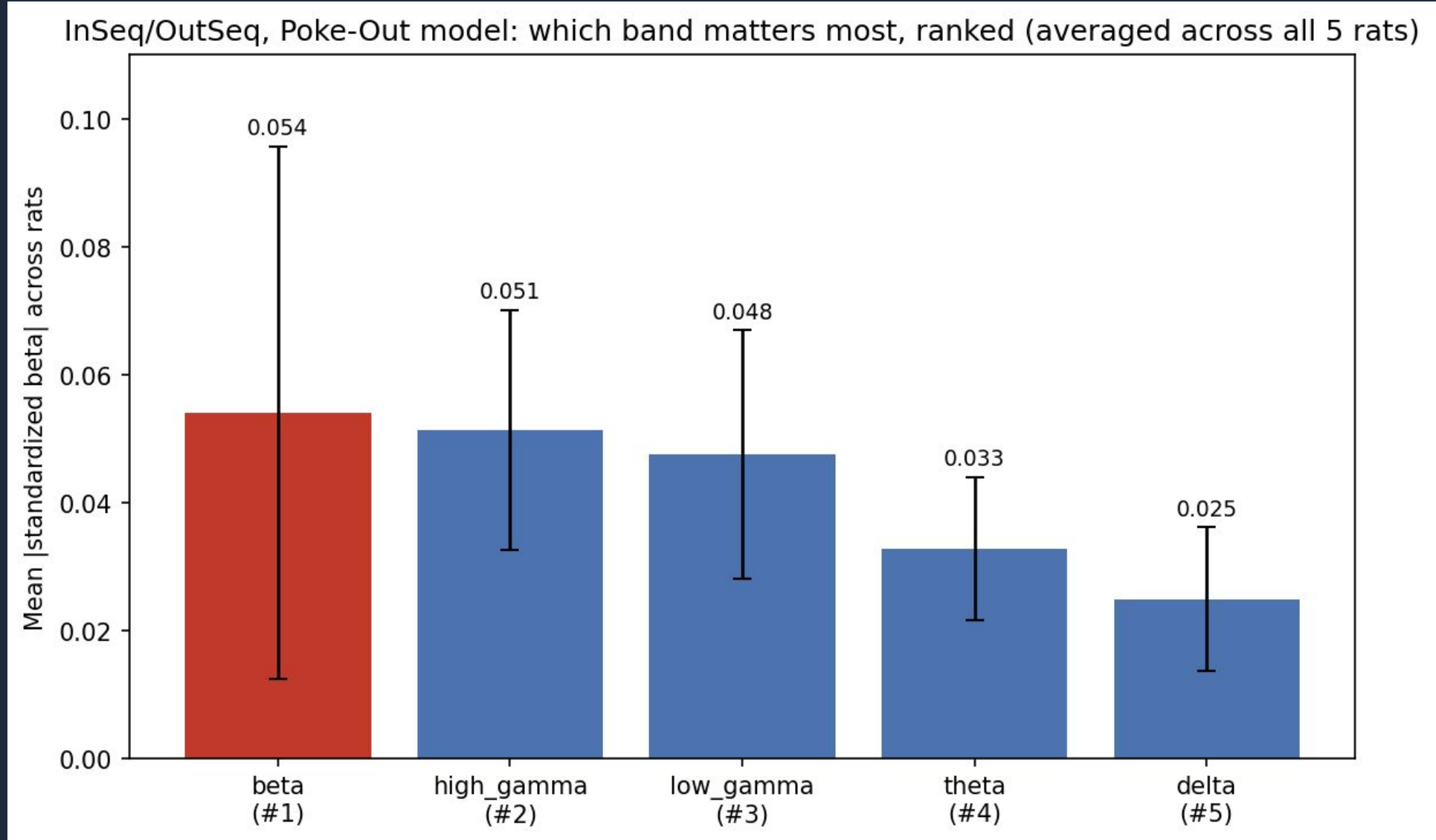
AUC & Decoding Performance: **0.808**



InSeq / OutSeq Decoding Performance – Poke-Out



InSeq/OutSeq Poke-Out Frequency Band



About the Experiment – Odor Identity Paradigm

01 / NEURAL CODING

Odor-Selective Neurons

Brain cells fire for specific scents while remaining quiet for others.

CELL RESPONSIVENESS

Selective Firing

02 / SENSORY SIGNAL

Unexpected Scents

Out-of-sequence odors trigger immediate firing to encode true odor identity.

SEQUENCE CONDITION

Non-Match Event

03 / BEHAVIORAL OUTPUT

Early Poke-Out

The subject detects sequence non-match and withdraws early to signal an error.

DECISION RESPONSE

Error Signaling

Source Dataset: Hippocampal ensembles represent sequential relationships among an extended sequence of nonspatial events

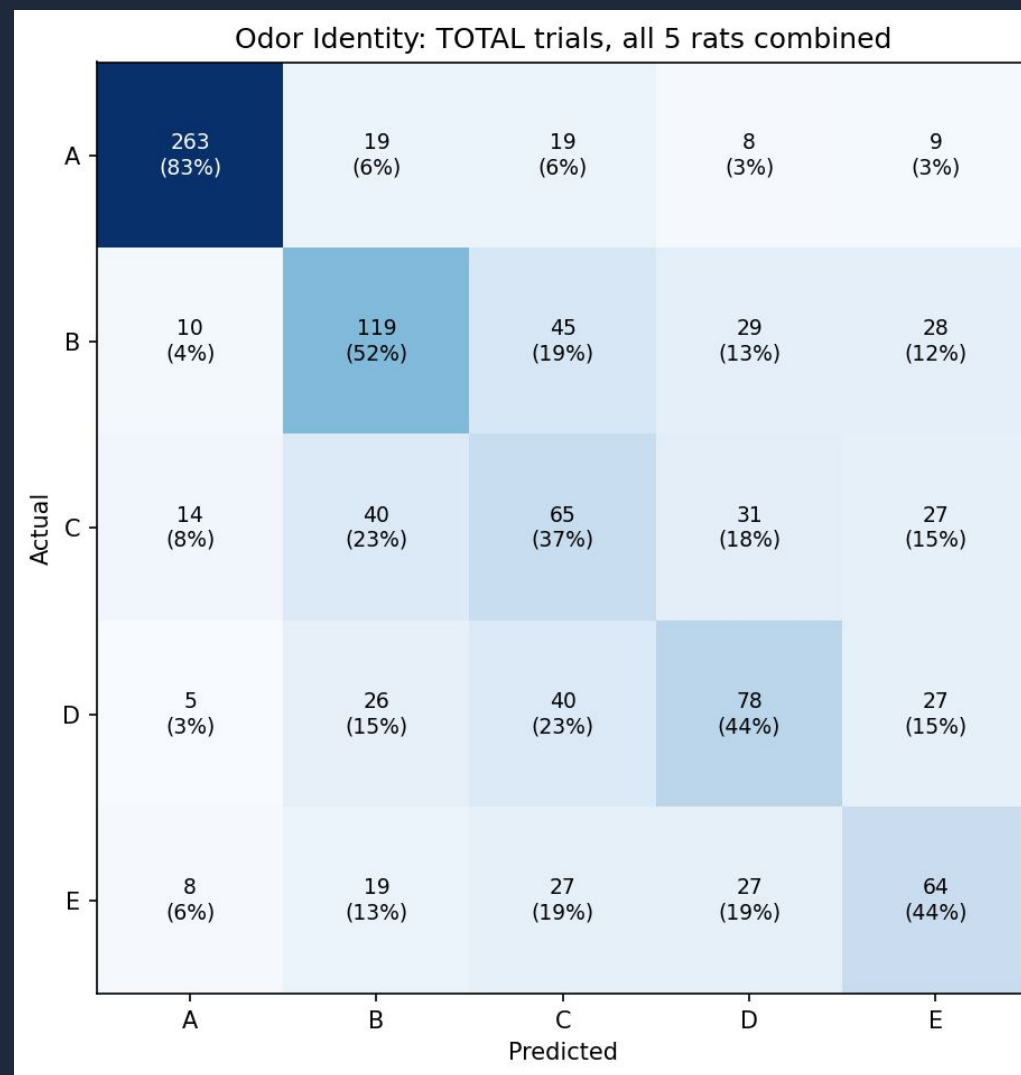
Odor Identity Decoding Performance

01. WINNING WINDOW & METRICS

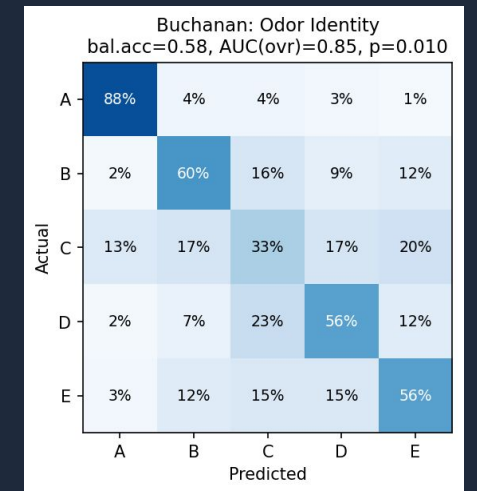
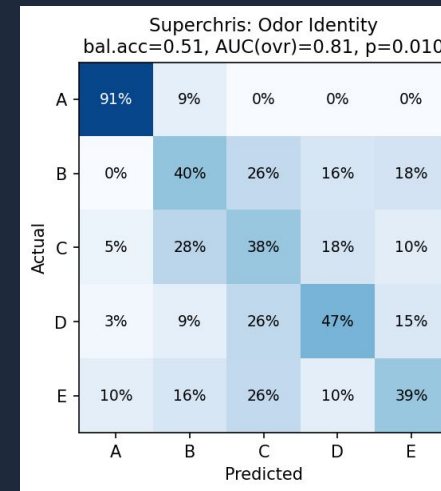
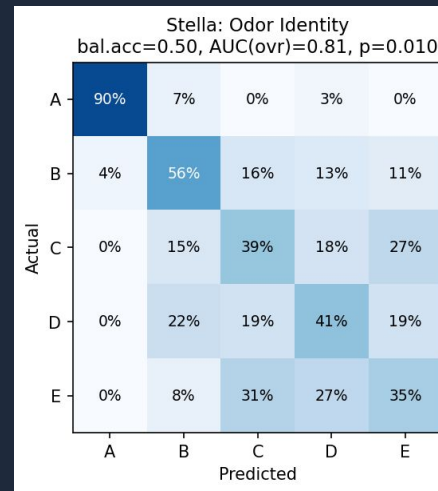
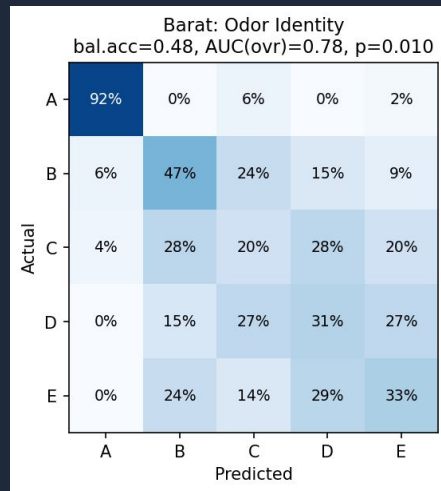
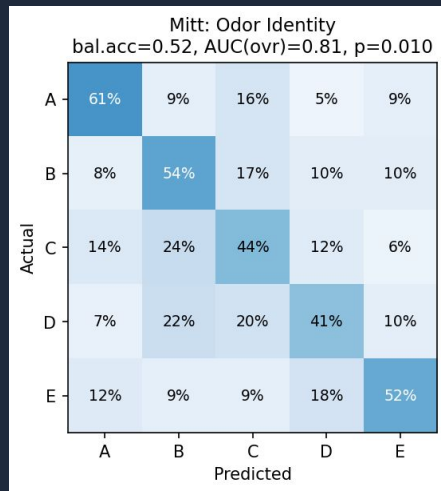
Odor Identity Analysis (0ms to 1250ms)

- 0 ms is the Poke-In event

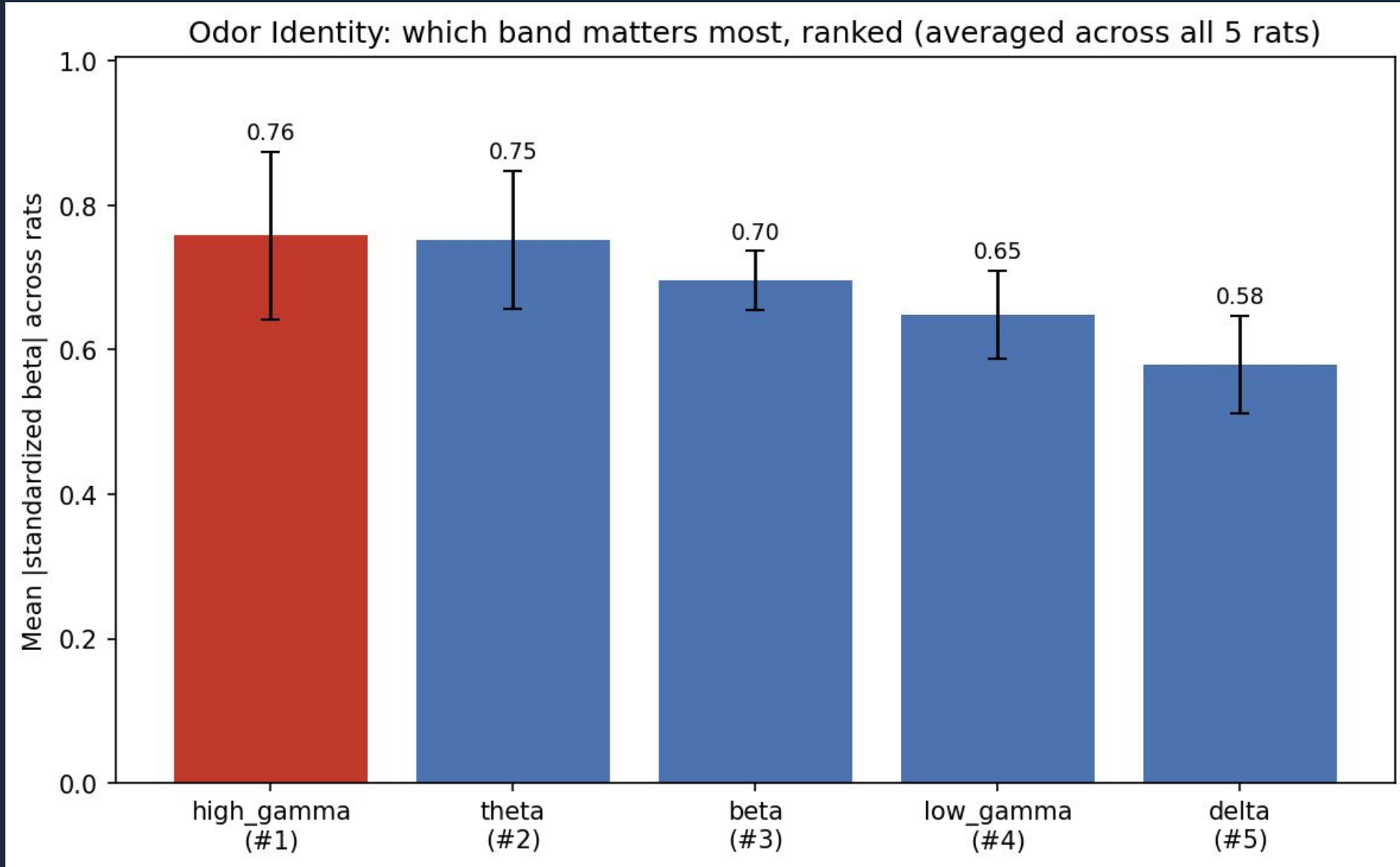
AUC & Decoding Performance: **0.813**



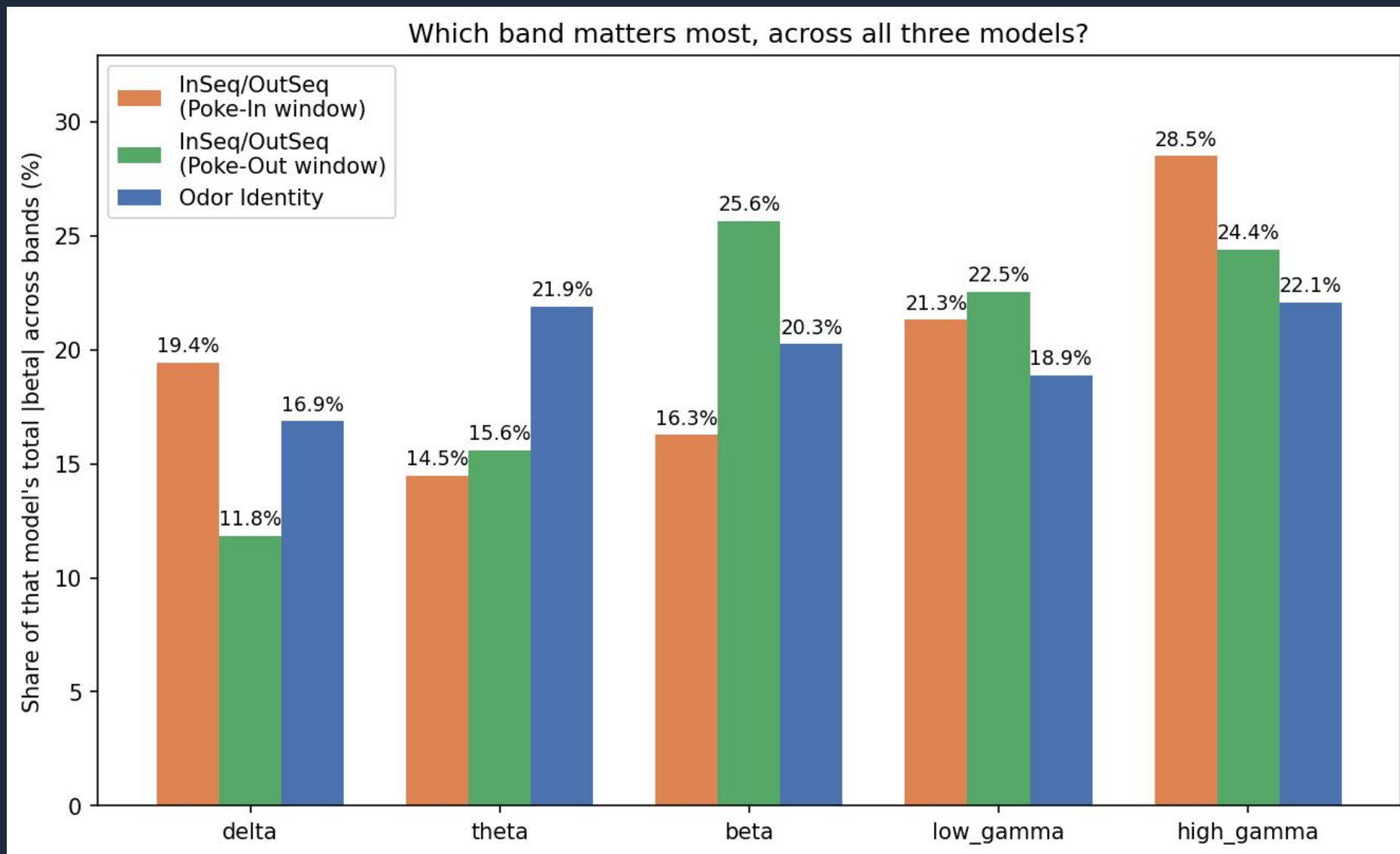
Odor Identity Decoding Performance



Odor Identity Frequency Band



Frequency Band Contributions to Sequence & Odor Decoding



Final Results

Next Steps

01

Variable Relationships

Conduct an investigation of the relationships between the different brainwaves.

02

Spectrogram Construction

Build and refine spectrograms.

03

Time Window Analysis

Focus detailed analysis on the specific time window from 250 ms to 500 ms.

Investigate critical transition dynamics and decoding changes within this epoch.